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The Incidence and the Risk Factors of Silent Embolic Cerebral Infarction After Coronary Angiography and Percutaneous Coronary Interventions

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Abstract

Silent embolic cerebral infarction (SECI) is a major complication of coronary angiography (CAG) and percutaneous coronary intervention (PCI). Patients with stable coronary artery disease (CAD) who underwent CAG with or without PCI were recruited. Cerebral diffusion-weighted magnetic resonance imaging was performed for SECI within 24 hours. Clinical and angiographic characteristics were compared between patients with and without SECI. Silent embolic cerebral infarction occurred in 12 (12%) of the 101 patients. Age, total cholesterol, SYNTAX score (SS), and coronary artery bypass history were greater in the SECI(+) group (65 ± 10 vs 58 ± 11 years, $P = .037$; 223 ± 85 vs 173 ± 80 mg/dL, $P = .048$; 30.1 ± 2 vs 15 ± 3, $P < .001$; 4 [33.3%] vs 3 [3.3%], $P = .005$). The SECI was more common in the PCI group (8/24 vs 4/77, $P = .01$). On subanalysis, the SS was significantly higher in the SECI(+) patients in both the CAG and the PCI groups (29.3 ± 1.9 vs 15 ± 3, $P < .01$; 30.5 ± 1.9 vs 15.1 ± 3.2, $P < .001$, respectively). The risk of SECI after CAG and PCI increases with the complexity of CAD (represented by the SS). The SS is a predictor of the risk of SECI, a complication that should be considered more often after CAG.

Keywords

silent embolic cerebral infarction, percutaneous coronary intervention, coronary angiography, SYNTAX score

Introduction

Coronary angiography (CAG) is the gold standard assessment for the evaluation of coronary artery disease (CAD). As an invasive diagnostic and therapeutic procedure, there are important periprocedural complications.[1] One of the well-known serious complication category is thromboembolic events including cerebrovascular events.[2] Atherosclerotic thromboembolism due to catheter manipulation, thrombus formation in the catheter lumen, and air embolism are the main causes of symptomatic or asymptomatic thromboembolic events.[1]

Silent embolic cerebral infarction (SECI) is defined as an embolic origin brain lesion, which is the result of vascular occlusion diagnosed incidentally by magnetic resonance imaging (MRI) or computed tomography (CT) in otherwise healthy participants or at autopsy.[3] Retrospective studies demonstrated that SECI was detected in 13% to 22% of cases undergoing diagnostic coronary angiography (CAG) and percutaneous coronary intervention (PCI).[4-6] As a noninnocent complication, SECI is associated with decline in cognitive function and increases the risk of dementia in the long term.[7] Diffusion-weighted magnetic resonance imaging (DW-MRI) is the most sensitive imaging technique in detecting acute cerebral infarction that appears as a high signal intensity within 24 hours.[8]

There are limited data regarding the development of SECI after diagnostic CAG and PCI. We aimed to evaluate the incidence and the predictors of the SECI in a heterogeneous group of patients undergoing CAG with or without PCI.

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Materials and Methods

Study Population

A total of 101 stable patients with CAD who underwent diagnostic CAG or PCI between November 2013 and December 2014 were prospectively included in this study. Acute coronary syndrome (unstable angina pectoris and myocardial infarction with or without ST-segment elevation), previous cerebrovascular event (ischemic or hemorrhagic), head trauma, malignancies related to the central nervous system, vascular and demyelinating disease of the central nervous system, significant occlusion in carotid and vertebral arteries, peripheral vascular disorders, atrial fibrillation, previous anticoagulant usage, MRI contraindications (cardiac pacemaker or claustrophobia), and chronic hepatic or renal failure were exclusion criteria in this study.

Neurological and general physical examinations were performed before and after CAG. Neurological deficits were quantified using the National Institutes of Health Stroke Scale.[9] All patients were neurologically intact before inclusion in this study. All patients included in this study gave their informed consent, and the study protocol was approved by the local ethics comittee.

Coronary Angiography

According to the standardized Judkins method, all patients underwent diagnostic CAG or PCI using a 6 to 7F catheter through femoral access. Acetylsalicylic acid of 100 mg was given to patients a week before intervention or a 300 mg loading dose just before intervention. Patients whose PCI continued after diagnostic CAG were administered 600 mg clopidogrel loading and 100 U/kg unfractioned heparin intravenously before starting the procedure. A water-soluble, nonionic and dimeric contrast agent, iodixanol (VisipaqueR 320 mg/mL, Amsterdam Health, Oslo, Norway), was used for all patients. Total fluoroscopy time was measured for all cases.

The severity of CAD was reported by 2 experienced cardiologists using the SYNTAX score (SS).[10] They were blinded to the clinical characteristics of the patients. To derive the SS, each coronary lesion >50% luminal obstruction in vessels with the diameter of $\geq$1.5 mm was separately scored and summed to provide the overall SS (www.Syntaxscore.com). The SS was calculated using dedicated software (version 2.11) that integrates the number of lesions with their specific weighting factors based on the amount of myocardium distal to the lesion and the morphologic features of each lesion.

Diffusion-Weighted MRI

A diffusion-weighted cranial MRI (DW-MRI) with a 3-T imaging unit was performed for all patients within 24 hours of coronary angiography. The MRI parameters for DW-MRI were a slice thickness of 5 mm, an interslice gap of 1 mm, 20 axial slices, and a field of view of 250 mm. The DW-MRI parameters included a repetition time of 6400 milliseconds, an echotime of 72.4 milliseconds, and a matrix number of 160 $\times$ 160. New revealed focal diffusion abnormalities detected on MRI after coronary angiography were considered as acute cytotoxic edema, so these findings were interpreted in favor of acute embolic infarction. All DW-MRI images were independently evaluated by 2 neurologists who are experienced in stroke and blinded to patients. Patients were divided into 2 groups: Cases with high signal intensity on DW-MRI were SECI(+) and cases with non-high signal intensity were SECI(−).

Statistical Analysis

All analyses were performed using SPSS 20 statistical software package (IBM SPSS Statistics). Categorical variables were expressed as numbers and percentages, whereas continuous variables were summarized as mean and standard deviation and as median and minimum–maximum where appropriate. The chi-square test was used to compare categorical variables between the groups. The normality of distribution for continuous variables was confirmed with the Kolmogorov-Smirnov test. For comparison of continuous variables between the 2 groups, the Student t test or Mann-Whitney U test was used depending on whether the statistical hypotheses were fulfilled or not. A receiver–operator characteristic (ROC) curve analysis was performed in order to identify the optimal cutoff point of SS for diagnosing SECI. A 2-tailed $P \leq .05$ was considered significant.

Results

Baseline Clinical and Angiographic Characteristics

Baseline demographic and CAG outcomes are shown in Table 1. Diagnostic CAG was performed in 77 (76%) cases, and PCI was performed in 24 (24%) cases. The SECI was detected in 12 (12%) of 101 cases (Figure 1). The general physical and neurological examinations of the patients before and after CAG and PCI revealed no pathological findings.

The CAD risk factors including gender, body mass index, smoking, hypertension, diabetes mellitus, hyperlipidemia, serum low-density lipoprotein cholesterol (LDL-C) levels, and left ventricular ejection fraction showed no difference in SECI(+) or SECI(−) groups. Mean age of the SECI(+) group was significantly greater than for the SECI(−) group (65 $\pm$ 10 vs 58 $\pm$ 11 years, respectively; $P = .037$). Although serum LDL-C and high-density lipoprotein cholesterol levels were similar in the 2 groups, total cholesterol levels were significantly higher in the SECI(+) group compared with the SECI(−) group (223 $\pm$ 85 vs 173 $\pm$ 80 mg/dL, $P = .048$). The SS was significantly greater in the SECI(+) group compared with the SECI(−) group (30.1 $\pm$ 2 vs 15 $\pm$ 3, $P < .001$).

The SECI was detected in 12 of all participants, 4 (33%) of 12 cases in the diagnostic CAG group and 8 (67%) of 12 cases in PCI group. The SECI was observed significantly more often in the intervention group (8/24 vs 4/77, $P = .01$). When the CAG and PCI groups were evaluated separately, there was no significant difference in fluoroscopy time between SECI(+) and SECI(−) groups (Table 2).

Table 1. Baseline Characteristics of the Patients.

	Total (n = 101)	SECI(+), n = 12 (12%)	SECI(−), n = 89 (88%)	*P* Value
Age	58 ± 11	65 ± 10	58 ± 11	.037
Sex, male/female	59/42	9/3	50/39	.350
Body mass index, kg/m^2	27.8 ± 2.4	27 ± 2	27.9 ± 2.5	.181
Hypertension	57 (56.4%)	7 (58.3%)	50 (56.2%)	.999
Diabetes mellitus	44 (43.6%)	7 (58.3%)	37 (41.6%)	.356
Hyperlipidemia	62 (61.4%)	7 (58.3%)	55 (61.8%)	.999
LDL	98 ± 45	105 ± 38	97 ± 46	.530
HDL	38 ± 7	36 ± 9	34 ± 5	.480
Triglyceride	239 ± 56	260 ± 54	236 ± 56	.164
Total cholesterol	179 ± 81	222 ± 85	173 ± 79	.048
LVEF	57.9 ± 8.9	53.3 ± 9.2	58.5 ± 8.7	.058
Smoker	59 (58.4%)	7 (58.3%)	52 (58.4%)	.999
SYNTAX score	16.8 ± 5.7	30.1 ± 2	15 ± 3	<.001
PCI / Diagnostic	24/77	8/4	16/73	.010
History of CABG	7 (6.9%)	4 (33.3%)	3 (3.3%)	.005

Abbreviations: LDL, low-density lipoprotein; HDL, high-density lipoprotein; LVEF, left ventricular ejection fraction; PCI, percutaneous coronary intervention; CABG, coronary artery bypass graft surgery; SECI, silent embolic cerebral infarction.

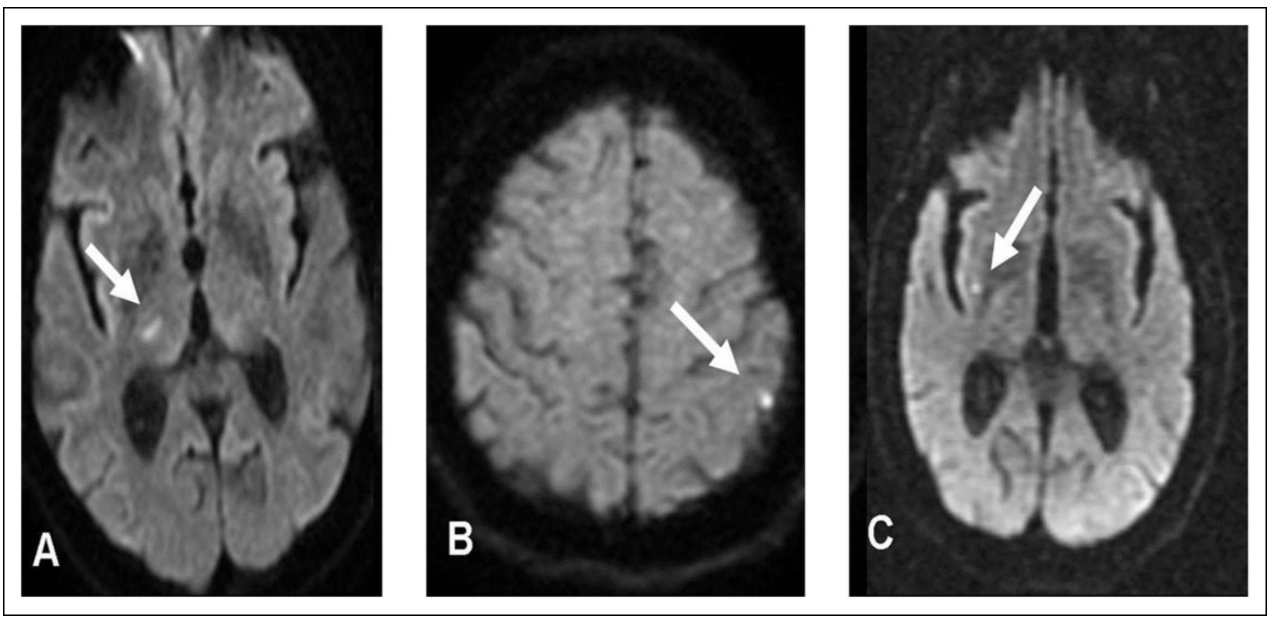

Figure 1. A, A band-like acute ischemic area in the right thalamic region with the diffusion-weighted cranial magnetic resonance imaging (DW-MRI) sequence . B, An acute ischemic area in the left postcentral gyrus with the diffusion-weighted cranial magnetic resonance imaging sequence. C, An acute ischemic area in the right perisylvian region with the DW-MRI sequence.

History of coronary artery bypass graft (CABG) was found in 3 patients in the diagnostic CAG group, and SECI was detected in one of these patients. History of CABG was found in 4 patients in the PCI group, and SECI was detected in 3 of these patients. Statistically significant difference was detected in terms of history of CABG between SECI(+) and SECI(−) groups. However, when we analyzed the CAG and PCI subgroups, there were no significant difference for history of CABG between SECI(+) and SECI(−) patients.

Factors that may affect development of SECI (gender, hypertension, diabetes mellitus, total cholesterol level, left ventricular ejection fraction, smoking, SS, PCI, and history of CABG) were evaluated by multivariate logistic regression analysis. In this model, independent predictors on developing SECI were SS, high total cholesterol levels, PCI, and CABG. When CAG and PCI groups were evaluated separately, the SS was only the predictor of SECI (Figure 2). The area under ROC curve for SS was 0.83. A SS cutoff point of 25.5 revealed a sensitivity and specificity of 90.9% and 91.1%, respectively.

Discussion

Coronary angiography is associated with periprocedural complications.[11] The SECI as a result of cerebral embolism

Table 2. Clinical Features of Patients in Subgroup of Diagnostic Coronary Angiography (CAG) and Percutaneous Coronary Intervention (PCI).

	Diagnostic Coronary Angiography				Percutaneous Coronary Intervention			
	Total (n = 77)	SECI(+), n = 4	SECI(−), n = 73	*P* Value	Total (n = 24)	SECI(+), n = 8	SECI(−), n = 16	*P* Value
Age	57 ± 11	60 ± 14	56 ± 11	.532	64 ± 12	67 ± 8	62 ± 13	.331
Sex, male/female	40/37	2/2	38/35	.999	19/5	7/1	12/4	.631
BMI	28.1 ± 2.3	27.5 ± 1.9	28.1 ± 2.3	.594	26.8 ± 2.6	26.6 ± 2.1	26.9 ± 2.9	.791
Hypertension	43 (55.8%)	3 (75%)	40 (54.8%)	.626	14 (58.3%)	4 (50%)	10 (62.5%)	.673
Diabetes Mellitus	30 (39%)	3 (75%)	27 (37%)	.293	14 (58.3%)	4 (50%)	10 (62.5%)	.673
Hyperlipidemia	47 (61%)	3 (75%)	44 (60.3%)	.999	15 (62.5%)	4 (50%)	11 (68.8%)	.412
LDL	92 ± 41	99 ± 31	92 ± 42	.727	115 ± 51	108 ± 42	118 ± 56	.671
HDL	35 ± 9	36 ± 9	35 ± 10	.455	33 ± 9	34 ± 10	33 ± 7	.511
Triglyceride	241 ± 55	255 ± 64	240 ± 55	.614	234 ± 58	263 ± 52	219 ± 56	.078
Total cholesterol	174 ± 81	222 ± 109	171 ± 80	.231	195 ± 80	223 ± 79	181 ± 79	.238
LVEF	59.7 ± 7.1	56.3 ± 11	59.9 ± 6.9	.318	52 ± 11.3	51.9 ± 8.5	52 ± 12.8	.490
Smoker	45 (58.4%)	3 (75%)	42 (57.5%)	0.637	14 (58.3%)	4 (50%)	10 (62.5%)	.673
SYNTAX score	15.8 ± 4.3	29.3 ± 1.9	15 ± 3	<.001	20.2 ± 7.9	30.5 ± 1.9	15.1 ± 3.2	<.001
History of CABG	3 (3.89%)	1 (25%)	2 (2.73%)	.133	4 (16.7%)	3 (37.5%)	1 (6.2%)	.091
Fluoroscopy time	6.7 ± 2.8	6.5 ± 2.4	6.7 ± 2.9	.903	10.3 ± 5.5	13.5 ± 6	8.8 ± 4.6	.172

Abbreviations: LDL, low-density lipoprotein; HDL, high-density lipoprotein; LVEF, left ventricular ejection fraction; CABG, coronary artery bypass grafting surgery; SECI, silent embolic cerebral infarction; BMI, body mass index.

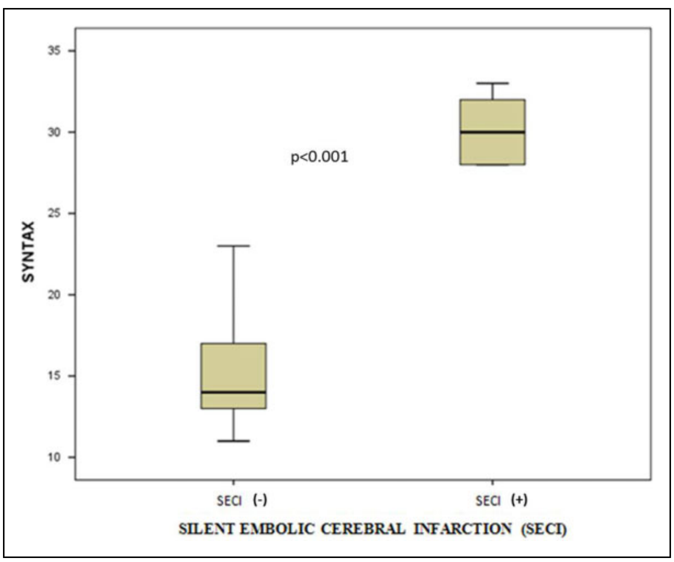

Figure 2. The relationship between silent embolic cerebral infarction (SECI) and SYNTAX Score (SS).

increases the risk of dementia and decline in cognitive function.[7]

Previous studies also reported that diagnostic catheterization procedures increased the incidence of SECI. In a retrospective study, the incidence of SECI was reported as 10.2% in the group of patients referred to CABG after CAG.[12] In this study, internal mammary artery (IMA) CAG was reported as the only independent factor for development of SECI. Many potential risk factors that lead to cerebral infarction after CAG have been described. Potential risk factors include multiple CAD, reduced ejection fraction, female gender, and left ventricular hypertrophy.[13]

The DW-MRI is a gold standard imaging method for determining SECI. In our study, postprocedural SECI was detected in 12% of our 101 patients. The SS is a common parameter for determining the complexity of CAD, and a higher SS is related to worse end points and major adverse coronary cardiac events.[10,14] In our study, the CAG and PCI groups were evaluated both together and separately. The SECI occurred significantly higher in patients with higher SS, and SS was the only predictor on occurrence of SECI in both the groups. Our finding may be attributable to patients with higher SS who had excessive atherosclerotic burden, calcification, and complexity not only existing in coronary vasculature; these pathological changes may also exist in aorta, carotid, and cerebrovascular arteries. Additionally, periprocedural difficulty depending on catheterization is higher in patients with high SS and also catheter manipulations are much more in this group of patients. When all these reasons are taken into consideration, aortic catheter manipulations may lead to damage in the aorta and increase the frequency of SECI. In another study, silent neuronal ischemia has been evaluated by neuron-specific enolase (NSE) levels after an intervention in patients with acute coronary syndrome.[14] In this study, a higher SS of patients who underwent CAG and PCI were related to higher NSE levels.

In our study, no significant correlation was detected between fluoroscopy time and frequency of SECI, and similarly no significant association was reported between fluoroscopy time and frequency of SECI development in other trials.[4,15]

The incidence of SECI increases in CAG after CABG.[16,17] Kim et al previously reported that IMA angiography increased the risk of SECI in patients who underwent CABG, and IMA CAG was an independent risk factor for SECI.[12] In our study population, SECI was detected in 1 patient after CAG who had CABG and in 3 patients after PCI who had CABG. In our study, when all patients were evaluated together, the incidence of SECI seemed to increase in patients who had CABG history.

However, the number of patients with CABG was low, and when the 2 groups were evaluated separately, no significant increase in frequency of SECI was detected in patients with CABG.

Our study has some limitations. First, the number of patients included was low. Second, patients had no radiological examinations for atherosclerosis in aorta, carotid, and vertebral arteries before CAG. Our study population was limited to the patients with stable CAD without other potential medical conditions that impact on development of SECI. Narrow lumen catheters such as 5F catheters were not used in our study (PCI was performed with 6-7F catheters). A radial approach was not used in our study. Association with CABG and incidence of SECI could not be assessed optimally due to the limited number of patients with previous CABG. Finally, our study was not a follow-up study so potential cognitive disorders and long-term effects on individuals associated with SECI could not be assessed.

Conclusions

The risk of SECI after CAG and PCI increases with the complexity of CAD (represented by the SS). The SS is a predictor of the risk of SECI, a complication that should be considered more often after CAG.

Authors' Note

Onur Sinan Deveci and Mustafa Demirtas contributed to conception and design, acquisition of data, drafting the article, revising, and final approval of the version to be published. Aziz Inan Celik, Firat Ikikardes, and Caglar Ozmen contributed to conception and design, acquisition of data, and drafting the article. Caglar Emre Caglıyan and Ali Deniz contributed to analysis and interpretation of data. Kenan Bicakci contributed to radiological analysis of data. Sebnem Bicakci, Ahmet Evlice, and Turgay Demir contributed to neurological analysis of data. Mehmet Kanadasi and Mesut Demir contributed to conception and design.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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